Geometry
Unit 9
Lesson 8 Homework

Name____Answer Key______
Date_____________________
Period____________________

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest thousandth if necessary.

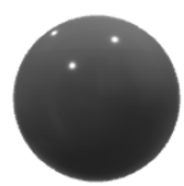
1. Find the surface area of a pyramid with a height of 9.8 meters, slant height of 10.34 meters, and a square base with a width of 6.7 meters.
$(6.7)^2 + \frac{1}{2}(10.34)(26.8) = 44.89 + 138.556 = 183.446$ m^2

2. Find the lateral area of a pyramid whose height is 9.8 meters, slant height is 10.34 meters, and whose base is a square with a width of 6.7 meters.
$LA = \frac{1}{2}(10.34)(26.8) = 138.556$ m^2

3. Find the lateral area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.
$LA = (177.8)(24.1) = 4284.98$ ft^2

4. Find the surface area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.
$2(31.8)(57.1) + (177.8)(24.1) = 7916.54$ ft^2

5. Find the surface area of a sphere with a diameter of 26.8 centimeters.
$SA = 4\pi(13.4)^2 = 718.24\pi \approx 2256.418$ cm^2 (when using the pi button)



6. Find the surface area of a hemisphere with a diameter of 102.9 millimeters.
$SA = 3\pi(51.45)^2 = 7941.3075\pi \approx 24{,}948.353$ mm^2 (when using the pi button)

7. The surface area of a sphere is 484π inches. What is the diameter of the sphere?

$484\pi = 4\pi r^2$

$121 = r^2$

$11 = r$

$d = 22$ in

8. Find the surface area of a cone with a radius of 16.7 yards, a height of 9.14 yards, and a slant height of 19.04 yards.

$SA = \pi(16.7)^2 + \pi(16.7)(19.04) = 278.89\pi + 317.968\pi = 596.858\pi$

≈ 1875.084 yd^2 (when using the pi button)

9. Find the lateral area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.

$LA = 2\pi(6.7)(9.4) = 125.96\pi \approx 395.715$ in^2 (when using the pi button)

10. Find the surface area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.

$SA = 2\pi(6.7)(9.4) + 2\pi(6.7)^2 = 125.96\pi + 89.78\pi = 215.74\pi \approx 677.767$ in^2 (when using the pi button)